GROUP 11 · BLUE OCEAN PITCH

HYPHA

Packaging that grows to fit.

Presented by

Matthew Carlin

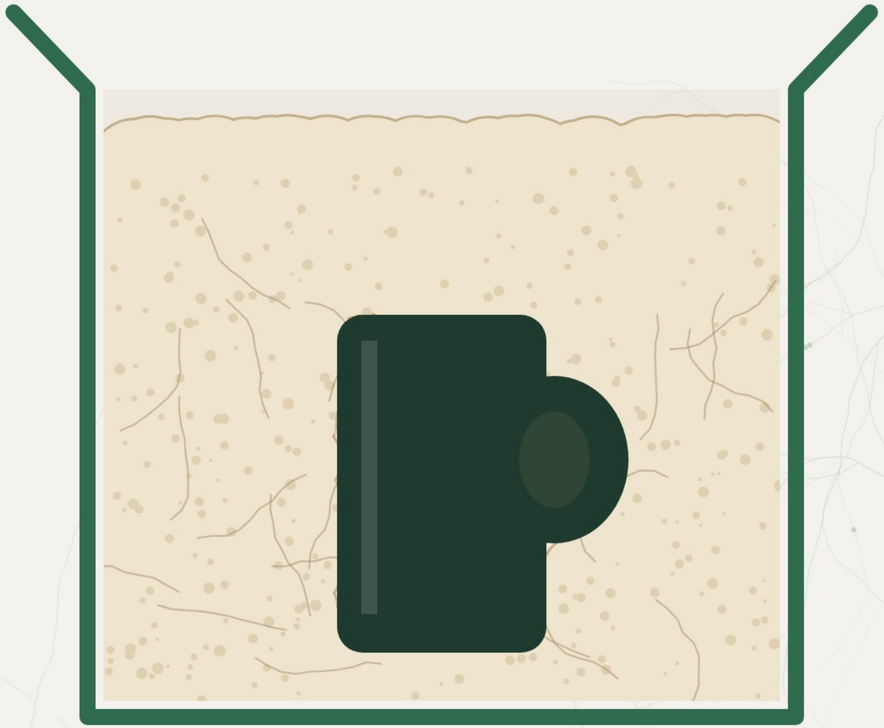
Daniel Thomas

Garrett McCormick

Sam Poole

Asher Lasky

Ledger

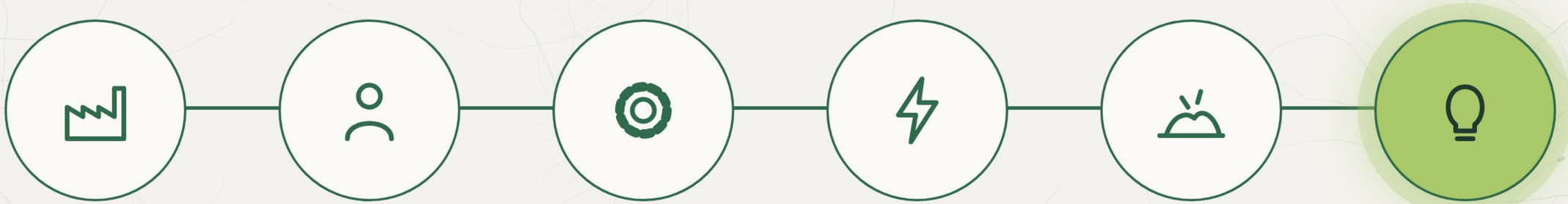


SET · READY TO SHIP

HYPHA PASTE

BEFORE WE BUILT ANYTHING

We stayed in the problem first.



Industry
E-commerce shipping

People
Pack stations, 3PLs

Process
Pillows, peanuts, paper

Friction
Generic fit, overfill

Consequence
Damage, landfill, new laws

Then: ideas
Only after the evidence

- Evidence before ideas
- Map the whole space
- Ask where it breaks

MIND MAP

The problem space of packaging

Packaging

Materials

<10%



of plastic packaging is **actually recycled**

Cost

+28%



green premium for recycled plastic (+20% for paper)

Waste

434.5Mt



of packaging waste a year, **projected**

Manufacturing

$50K+



for **one custom mold**, 12 to 24 weeks to make

Customer

73%



say **premium packaging** means a better brand

Sources: Materials [5][6][7] · Cost [10] · Waste [9] · Customer [11] · Mold cost and lead time from team research. Full evidence in the speaker notes; references on the last pages.

THE PROBLEM

Today, the empty space gets stuffed.

Plastic air pillows | Foam peanuts | Crumpled paper

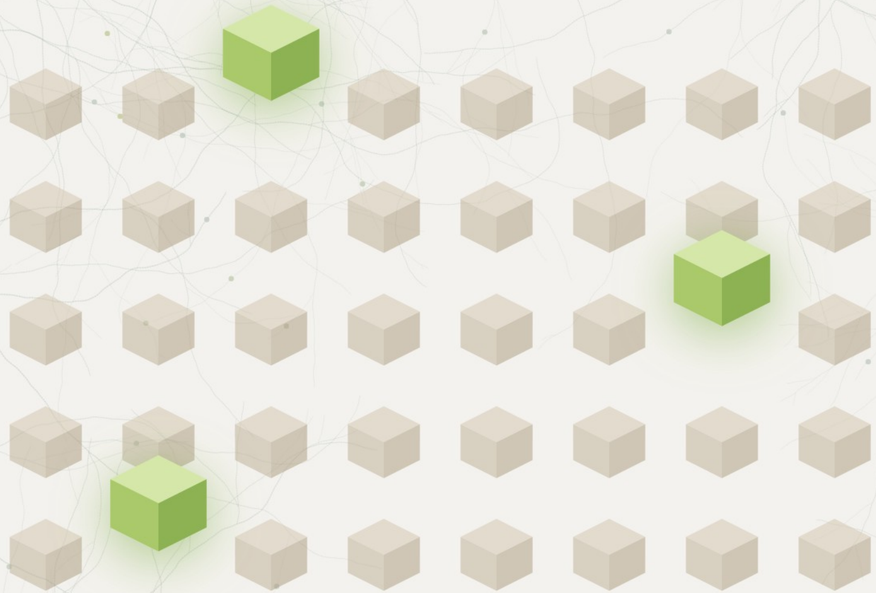
- **Generic fit**
  Products shift and break in transit
- **Rarely recycled**
  Most of it ends up in landfill
- **Under new laws**
  EPR and polystyrene bans target it



Sources: plastic waste and recycling [5][6][7] · EPR and polystyrene bans [1] · Full references on the last pages.

THE PROBLEM · WASTE

Packaging is used once, then buried.



1 box = 1% of US plastic waste

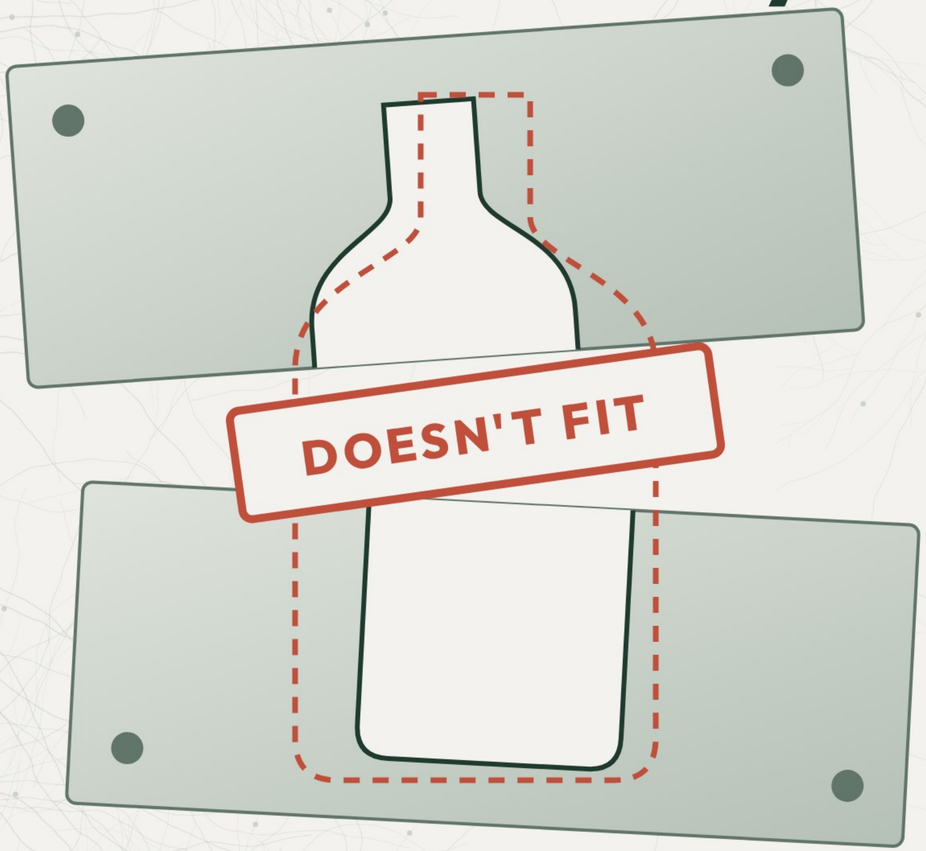
<10%

of that packaging is recycled.
The rest is buried or burned.

Sources: [5] US EPA, Plastics and Containers & Packaging data · [6] Our World in Data, "Packaging is the source of 40% of the planet's plastic waste" · [7] Beyond Plastics · [8] The Recycling Partnership

THE OTHER OPTION · CUSTOM INSERTS

A custom fit today means a new mold.



ONE CUSTOM MOLD

$200K+

starting around $50K

BEFORE THE FIRST PART SHIPS

24 weeks

0 — 12 · best case — 24

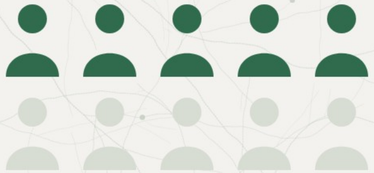
×2 Product changes? Pay it all again.

Mold cost and lead time from team research.

THE PROBLEM · CUSTOMER

The box is the brand's first handshake.

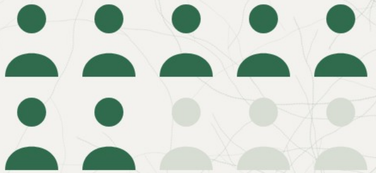
53%



came back after a great unboxing



73%



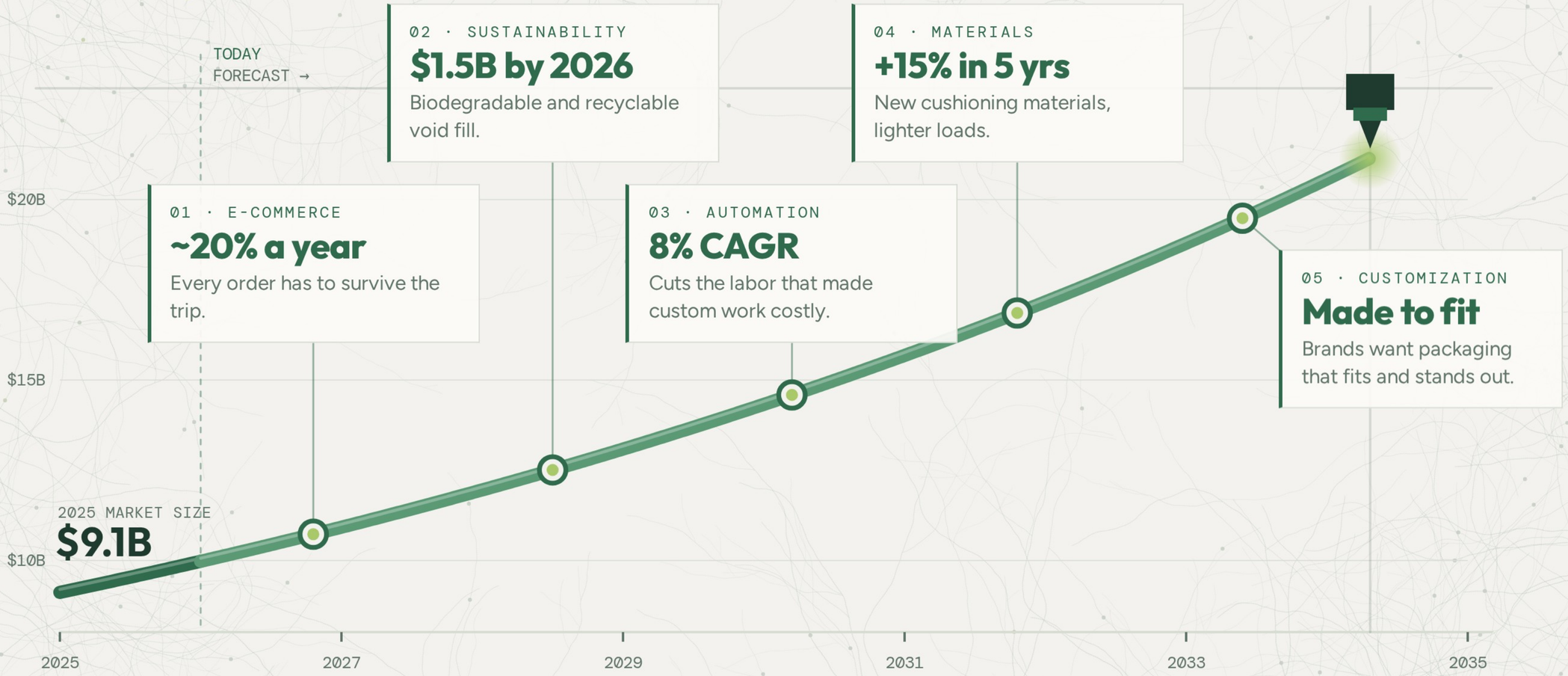
judge the brand by its packaging

It has to arrive intact. Protection is why shoppers pick boxes over mailers.

Sources: [13] 2025 study of 1,193 consumers · [14] Journal of Consumer Psychology unboxing experiments · [11] Adobe Express packaging survey · [12] Ryder e-commerce study via Packaging Technology Today

WHY NOW · US & CANADA PROTECTIVE PACKAGING

A $9.1B market, compounding.



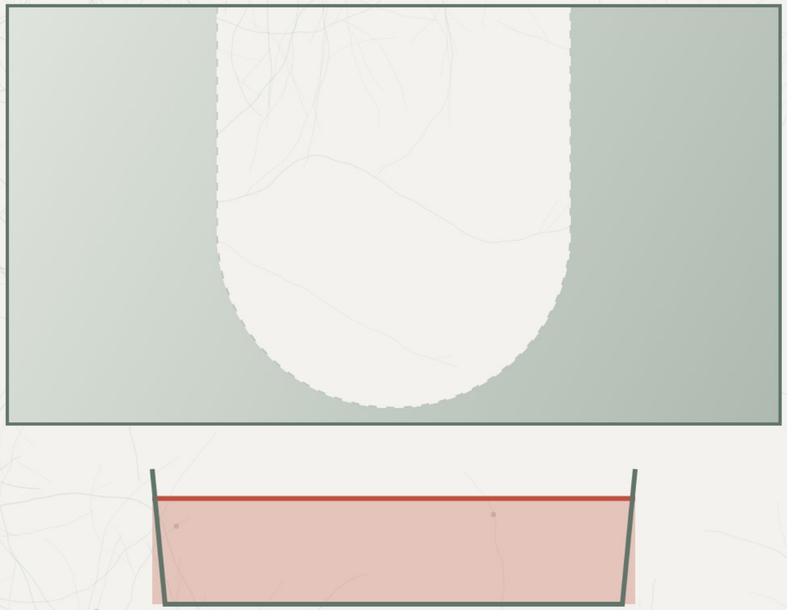
Market size: [1] Future Market Insights, US & Canada Protective Packaging Market (2025, 2035 forecast, 5.1% CAGR as reported). Drivers: [4] Market Research Future, Void Fill Packaging System Market. Curve between the two endpoints is interpolated.

THE EXPONENTIAL DRIVER WE'RE BETTING ON

Put material only where it's needed.

Cut it out

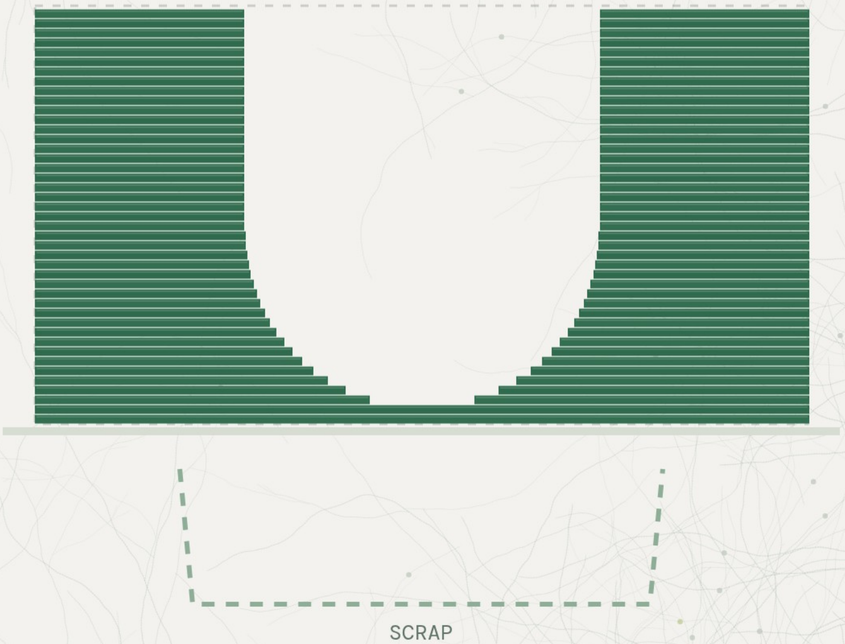
TRADITIONAL



40% of the block becomes scrap

Build it up

ADDITIVE



0% scrap, and no mold

Illustration: the scrap share is measured from this drawing, not an industry figure.

PRODUCT DEFINITION · HOW IT WORKS

Scan. Print. Expand. Return to soil.

A smart void-fill system that 3D-prints biodegradable, mycelium-based foam into the box for a custom fit.

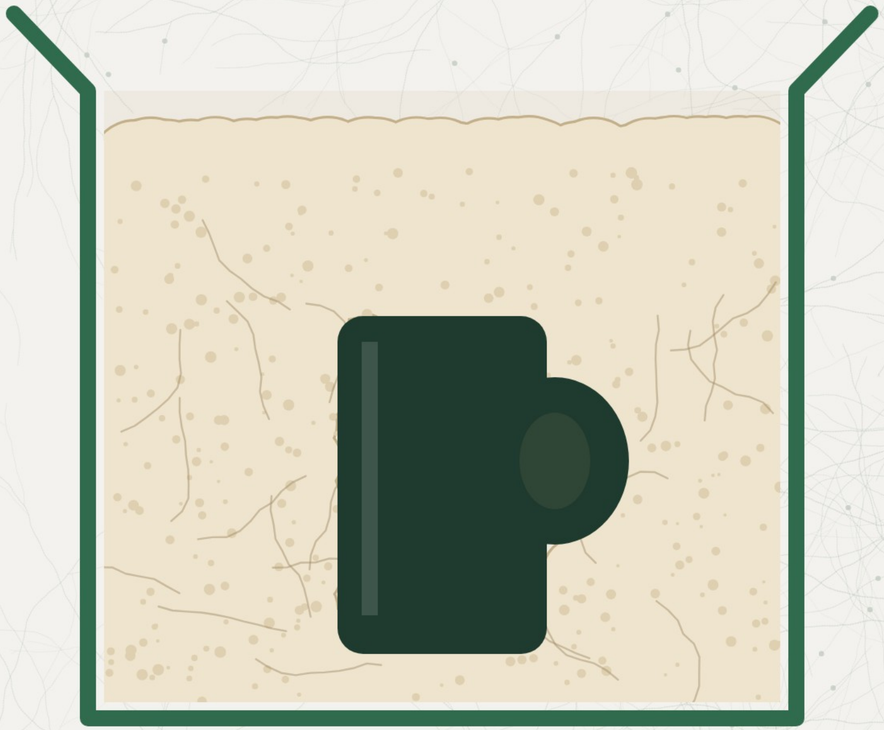
Scan
A camera maps the box and product in 3D

Print
A nozzle traces only the empty space

Expand
The paste foams and locks the product in

Return to soil
Composts at home in weeks

CONCEPT RENDER



EXPANDING

HYPHA PASTE

THE MATERIAL

Mushroom roots + farm waste.

Mycelium
the root network of mushrooms

+

Hemp hurd
agricultural waste fiber

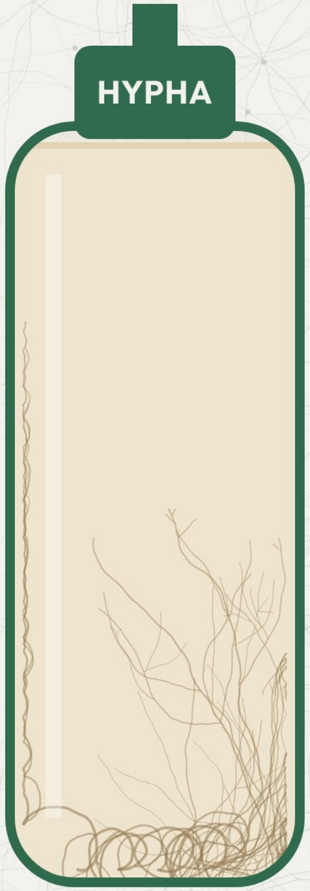
+

Corn husk
agricultural waste fiber

Home-compostable

Abundant & renewable

Refillable cartridges



REFILLABLE CARTRIDGE

COMPETITIVE ANALYSIS · SWOT

Where we're strong, and where we're exposed.

Strengths INTERNAL · HELPFUL

- The only option we found that is both made on demand and home compostable [16]-[22]
- Custom fit for every box with no molds; molded mycelium needs $2K–$9K tooling [20]
- Plastic-free material: mushroom roots plus farm waste

Weaknesses INTERNAL · HARMFUL

- No prototype yet: expansion, set time and strength are untested
- Cost per order unknown; Holiday pays ~$1 an order and caps a switch at 2× [15]
- Mycelium grows slowly (7 days for molded parts), so the paste must be pre-grown [20]

Opportunities EXTERNAL · HELPFUL

- US & Canada void fill: $1.9B in 2025, growing to $3.2B by 2035 [1]
- E-commerce is the largest use, at 42.1% of all void fill [2]
- EPR laws (CA, OR, WA) and California's 25% single-use plastic cut by 2032 push brands off plastic [1] [23]

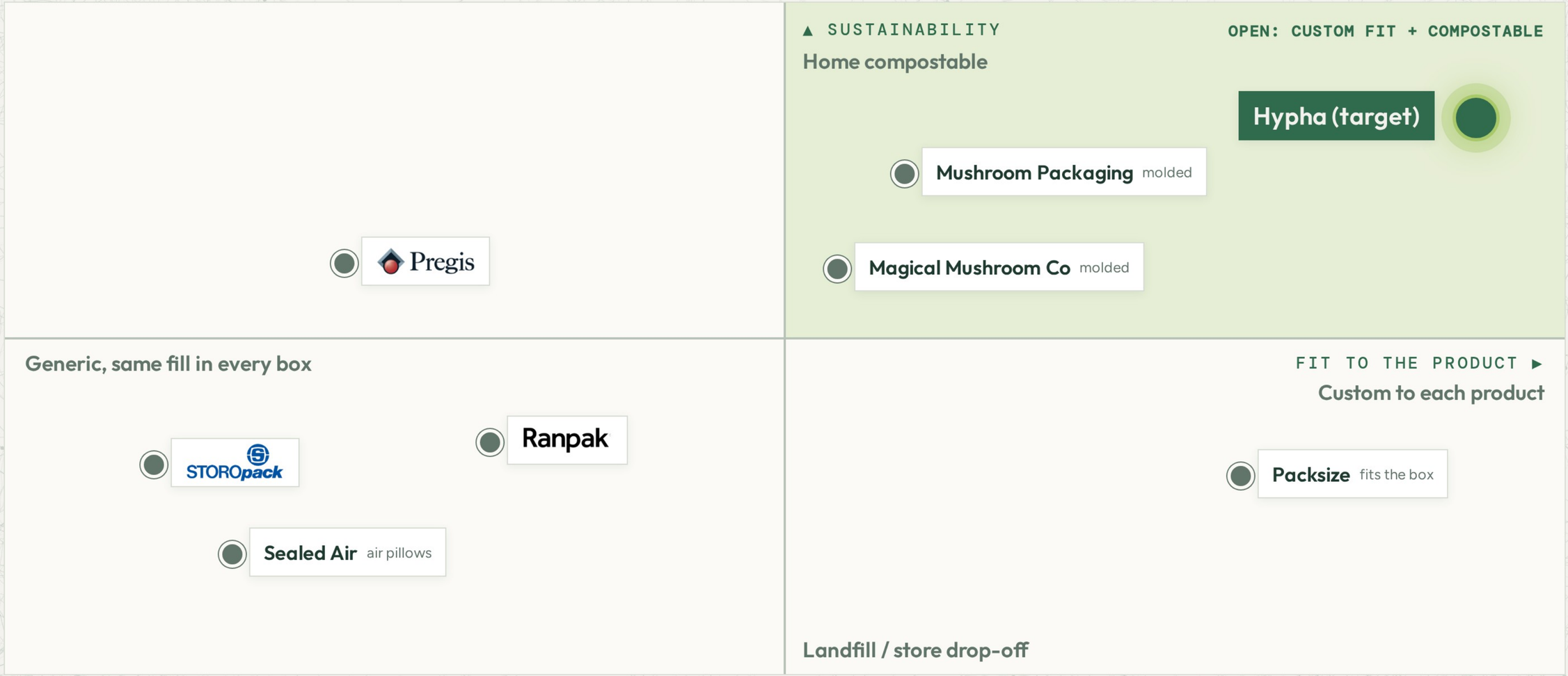
Threats EXTERNAL · HARMFUL

- Incumbents already run fill machines at the pack line; the top 3 hold 18% of the market [3] [16] [17]
- Ranpak sizes paper fill with AI vision; Packsize's right-sized boxes cut the need for fill [18] [22]
- Paper fill is cheap and curbside recyclable, "green enough" for many buyers [17] [18]

Sources: [1] Future Market Insights, US & Canada Protective Packaging · [2] Market Intelo, Void Fill Packaging · [3] FMI, Void Fill Systems Market Share · [15] Holiday interview (Group 11) · [16] Sealed Air · [17] Pregis · [18] Ranpak · [20] Mushroom Packaging · [22] Packsize · [23] Grand View Research

COMPETITIVE ANALYSIS · LANDSCAPE

Sustainability vs. fit: one corner is empty.



Placed from each company's own product page: [16] Sealed Air · [17] Pregis · [18] Ranpak · [19] Storopack · [20] Mushroom Packaging · [21] Magical Mushroom Co · [22] Packsize. Hypha's spot is a design target. Instapak and Cruz Foam are not placed: no product data in our sources.

COMPETITIVE ANALYSIS · COMPARISON CHART

Only Hypha checks every box.

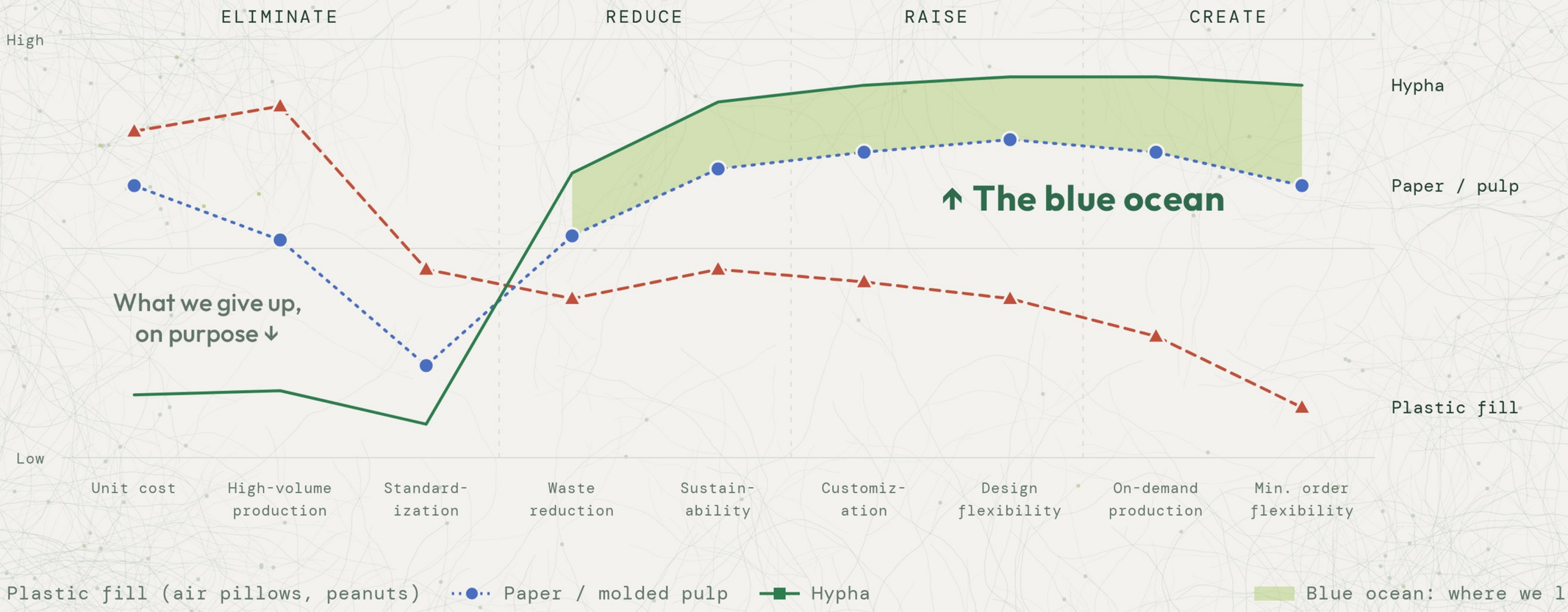
	HYPHA TARGET	Sealed Air	Pregis	Ranpak	STOROpack	Mushroom Packaging	Magical Mushroom Company	Packsize
Custom fit to each product	✓	✗	✗	✗	✗	✓	?	✓
Home compostable	✓	✗	✗	✗	✗	✓	?	✗
Made on demand at the pack line	✓	✓	✓	✓	✗	✗	✗	✓
No molds or tooling	✓	✓	✓	✓	✓	✗	?	✓
Plastic-free	✓	✗	✗	✓	✓	✓	✓	✓

✓ Yes ✗ No ? Not stated Logos link to each company's page.

From each company's own page: Sealed Air [16] · Pregis [17] · Ranpak [18] · Storopack [19] · Mushroom Packaging [20] · Magical Mushroom Co [21] · Packsize [22]. Mushroom Packaging fits by molding each shape ($2K-$9K tooling, 7-day grow); Packsize fits the box, not the fill; Pregis Bio One is industrially, not home, compostable. Hypha's column is design targets.

BLUE OCEAN VALUE MAP

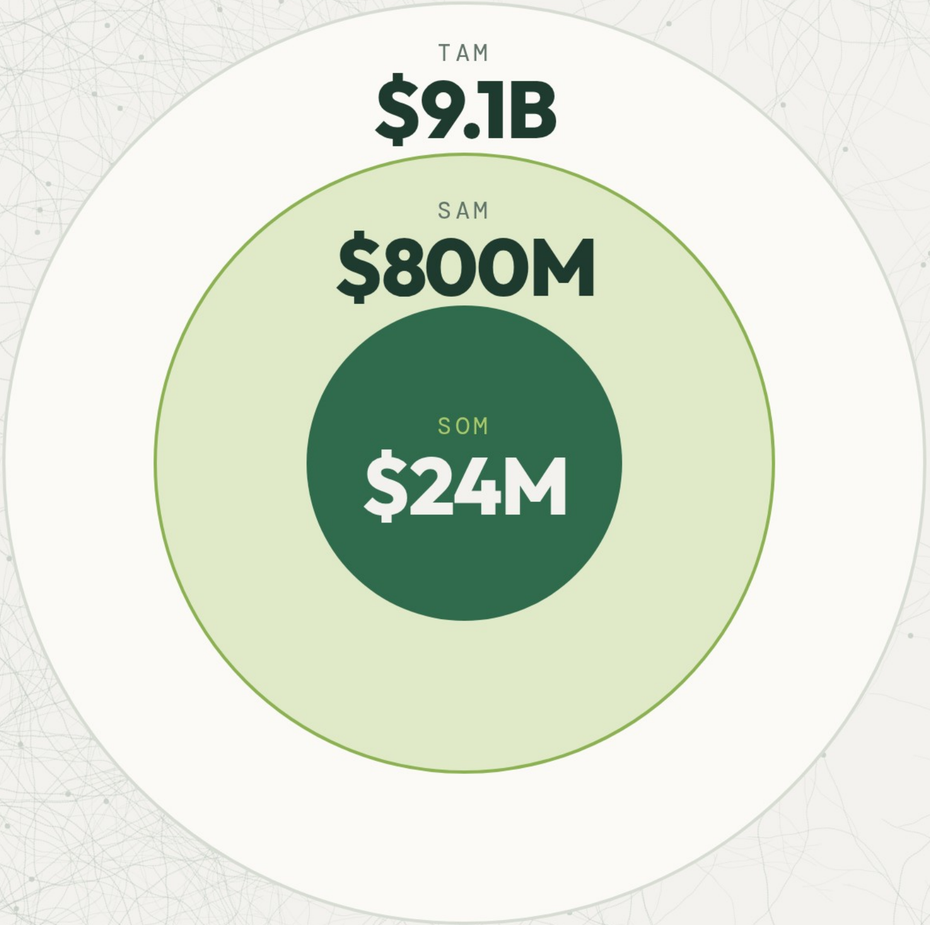
Stop competing on volume. Compete on fit.



Scores are our team's estimates, to be checked in customer interviews.

MARKET SIZE

From $9.1B to our first $24M.



TAM

$9.1B

US & Canada protective packaging, 2025

SAM

$800M

E-commerce void fill

$1.9B void fill × 42.1% e-commerce =

$800M

SOM

$24M

3% of SAM, benchmarked against today's players

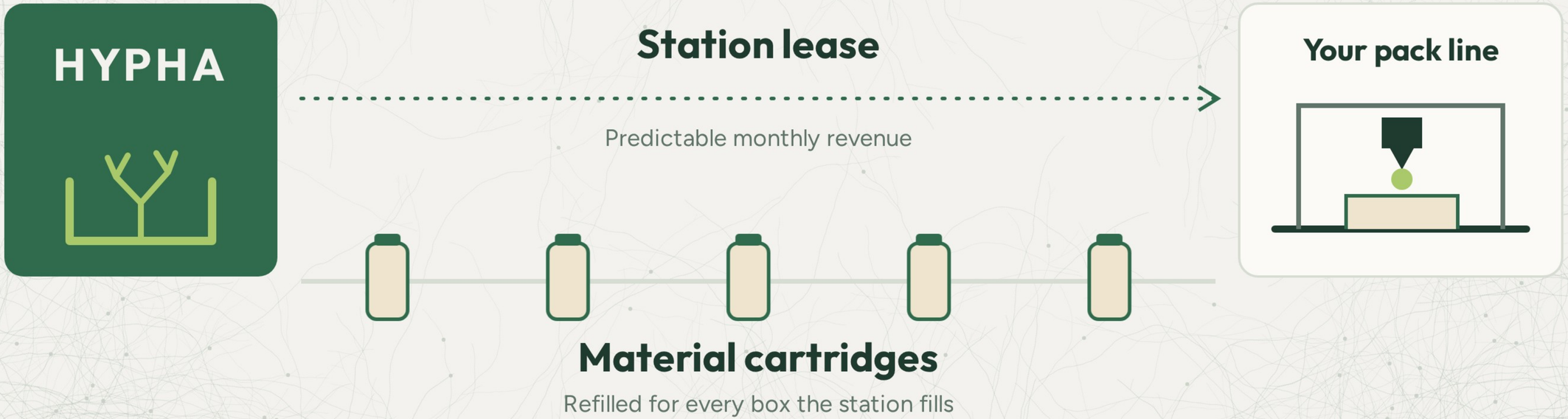
Top 10 together	36%
Top 3 together	18%
Each top-3 player	~6%
Hypha target	3%

Stretch case: 5% of SAM gives about $40M.

Circles not to scale. Sources: [1] Future Market Insights, US & Canada Protective Packaging Market · [2] Market Intelo, Void Fill Packaging Market (e-commerce share) · [3] Future Market Insights, Void Fill Packaging Systems Market Share Analysis. SOM share is our assumption.

BUSINESS MODEL

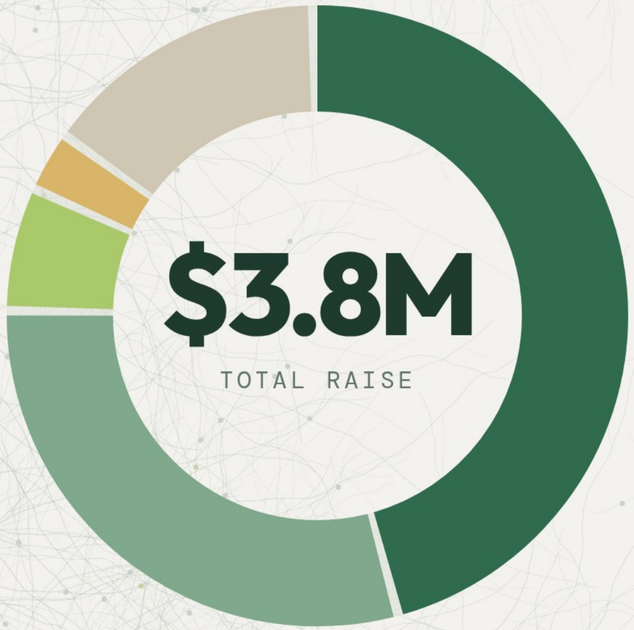
Lease the station. Sell the cartridges.



Like a printer and its ink: revenue grows with every box shipped.

USE OF FUNDS

$3.8M takes Hypha from lab to cash-flow positive.



Station builds First ~70 stations at $25K each	**$1.75M**	46%
Team Engineers, ops, first sales hire	**$1.12M**	30%
Sales + marketing Pilot brands, trade shows	**$245K**	6%
R&D + certification Material tests, home-compost certification	**$120K**	3%
Operations + reserve Rent, software, ~15% buffer	**$565K**	15%
Total	**$3.8M**	100%

View our Pro Forma →

WHAT THE MONEY UNLOCKS

1 · FIRST
Prove the material
Fill sets in seconds and composts at home

2 · NEXT
First pilot station live
One brand, real orders, at or under $2 a box

3 · MONTH 20
EBITDA positive
About 41 stations running

4 · MONTH 34
Cash-flow positive
About 210 stations running

Source: [24] Hypha Pro Forma (Google Sheet, linked on the slide). Raise = peak cash need of $3.32M (month 33) plus a ~15% buffer. Each bucket is its share of spending until cash flow turns positive, scaled to $3.32M; the buffer sits in Operations + reserve. Costs and volumes are team assumptions.

VALUE PROPOSITION · GEOFF MOORE TEMPLATE

How we position Hypha.

For
TARGET CUSTOMER

direct-to-consumer brands shipping fragile or oddly shaped products

who
NEED PROBLEM

want to get plastic out of their packaging without risking broken orders or a cheap-looking unboxing,

Hypha is
PRODUCT CATEGORY

an on-demand, compostable void-fill system

that
BENEFIT SOLUTION

replaces plastic fill with home-compostable foam made from mushroom roots and farm waste, printed to fit each product so it stays locked in place.

Unlike
COMPETITION

air pillows and paper fill, which are the same in every box, or molded mushroom packaging, which needs $2K–$9K tooling and a 7-day grow, Hypha needs no plastic and no molds.

Template: Geoff Moore, *Crossing the Chasm*. Competitor facts: [16] Sealed Air · [18] Ranpak · [20] Mushroom Packaging. Customer need from our Holiday interview [15]. Hypha's benefits are design targets.

HYPHA

Protection that fits. A future with less plastic.

A mycelium-based packaging concept designed to expand around products and replace conventional void fill.



Concept visualization

01 **CUSTOM FIT**
Designed to conform around different product shapes.

02 **LESS PLASTIC**
A proposed alternative to plastic foam and fillers.

03 **BETTER END OF LIFE**
Aiming for compostable protection after delivery.

CONCEPT STAGE · Expansion, protective performance and compostability require validation.

OUR VALUE PROPOSITION

Packaging shouldn’t outlive the order.

Every Hypha fill is built on one belief: **what protects a product for a few days shouldn’t sit in a landfill for years.**
So we print each fill from mushroom roots and farm waste, **shaped to the product and made to return to the soil.**

REFERENCES

Sources (1 of 3)

[1] Future Market Insights, Protective Packaging Industry Analysis in United States and Canada, 2025 to 2035 (TAM $9.1B; void fill $1.9B)
https://www.futuremarketinsights.com/reports/united-states-and-canada-protective-packaging-market

[2] Market Intelo, Void Fill Packaging Market Research Report 2034 (e-commerce = 42.1% of void fill, 2025)
https://marketintelo.com/report/void-fill-packaging-market

[3] Future Market Insights, Void Fill Packaging Systems Market Share Analysis (top 3 = 18%, top 10 = 36%)
https://www.futuremarketinsights.com/reports/void-fill-packaging-systems-market-share-analysis

[4] Market Research Future, Void Fill Packaging System Market (growth drivers)
https://www.marketresearchfuture.com/reports/void-fill-packaging-system-market-38467

[5] US EPA, Facts and Figures about Materials, Waste and Recycling: Plastics; Containers and Packaging
https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/plastics-material-specific-data

https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/containers-and-packaging-product-specific

[6] Our World in Data, "Packaging is the source of 40% of the planet's plastic waste"
https://ourworldindata.org/data-insights/packaging-is-the-source-of-40-of-the-planets-plastic-waste

[7] Beyond Plastics, The Real Truth About the U.S. Plastics Recycling Rate
https://www.beyondplastics.org/publications/us-plastics-recycling-rate

[8] The Recycling Partnership, State of Recycling: The Present and Future of Residential Recycling in the U.S. (2024)
https://recyclingpartnership.org/wp-content/uploads/dlm_uploads/2024/01/Recycling-Partnership-State-of-Recycling-Report-1.12.24.pdf

REFERENCES

Sources (2 of 3)

[9] Smithers, "Packaging waste outpacing global capacity for sustainable management" (434.5M tonnes, 2025)
https://www.smithers.com/resources/2025/december/packaging-waste-outpacing-global-capacity

[10] Packaging Technology Today, "The Rise of Sustainable Packaging in the U.S.: Trends, Drivers, and the Road Ahead"
https://www.packagingtechtoday.com/materials/the-rise-of-sustainable-packaging-in-the-u-s-trends-drivers-and-the-road-ahead/

[11] Adobe Express, "The power of the package" consumer survey (73% say premium packaging improves view of brand)
https://www.adobe.com/express/learn/blog/power-of-the-package

[12] Packaging Technology Today, "Study Finds E-commerce Packaging Choices Impact Purchasing Decisions" (Ryder study; boxes chosen for protection)
https://www.packagingtechtoday.com/featureds/study-finds-e-commerce-packaging-choices-impact-purchasing-decisions/

[13] 2025 consumer study of 1,193 shoppers (unboxing and repeat purchase)
Cited by the team; link not yet added.

[14] Journal of Consumer Psychology, unboxing experiments
Cited by the team; link not yet added.

[15] Customer interview: Holiday (DTC apparel brand) team, conducted by Ledger, Group 11. Primary research.
Interview notes held by the team.

[16] Sealed Air, Fill-Air Extreme inflatable void fill (NASSCO distributor listing)
https://www.nasscoinc.com/item/100699531/Sealed-Air/

REFERENCES

Sources (3 of 3)

[17] Pregis, Void Fill Packaging (EasyPack paper, AirSpeed air cushions, Bio One loosefill)
https://uk.pregis.com/eu-solutions/by-need/inside-the-box-protection/void-fill-packaging/

[18] Ranpak, Void Fill (FillPak paper systems, DecisionTower AI-sized fill)
https://www.ranpak.com/void-fill/

[19] Storopack, PAPERbubble paper padding
https://www.storopack.us/products/flexible-protective-packaging/paper-padding/paperbubbler/

[20] Mushroom Packaging, a division of Ecovative Design (7-day grow, $2K-$9K tooling, $1,350 prototyping, home compostable)
https://mushroompackaging.com/

[21] Magical Mushroom Company (mycelium packaging, UK)
https://magicalmushroom.com/

[22] Packsize, FAQ (on-demand right-sized boxes)
https://www.packsize.com/resources/faq

[23] Grand View Research, Sustainable Packaging Market Report (California single-use plastic law)
https://www.grandviewresearch.com/industry-analysis/sustainable-packaging-market-report

[24] Hypha Pro Forma, team financial model (Google Sheet): revenue, costs, EBITDA and cash need by month
https://docs.google.com/spreadsheets/d/1l65aVK5H2IcMJ-W-V3p3NfhqUJOvrHgzxo5abZoMqqc/edit?usp=sharing

Notes: Calculated figures (21% void fill share, $800M SAM, $24M SOM) are derived from [1] to [3]. The SOM capture rate (3%) is a team assumption benchmarked against incumbent shares. Value-map scores and the additive-manufacturing scrap illustration are team estimates.